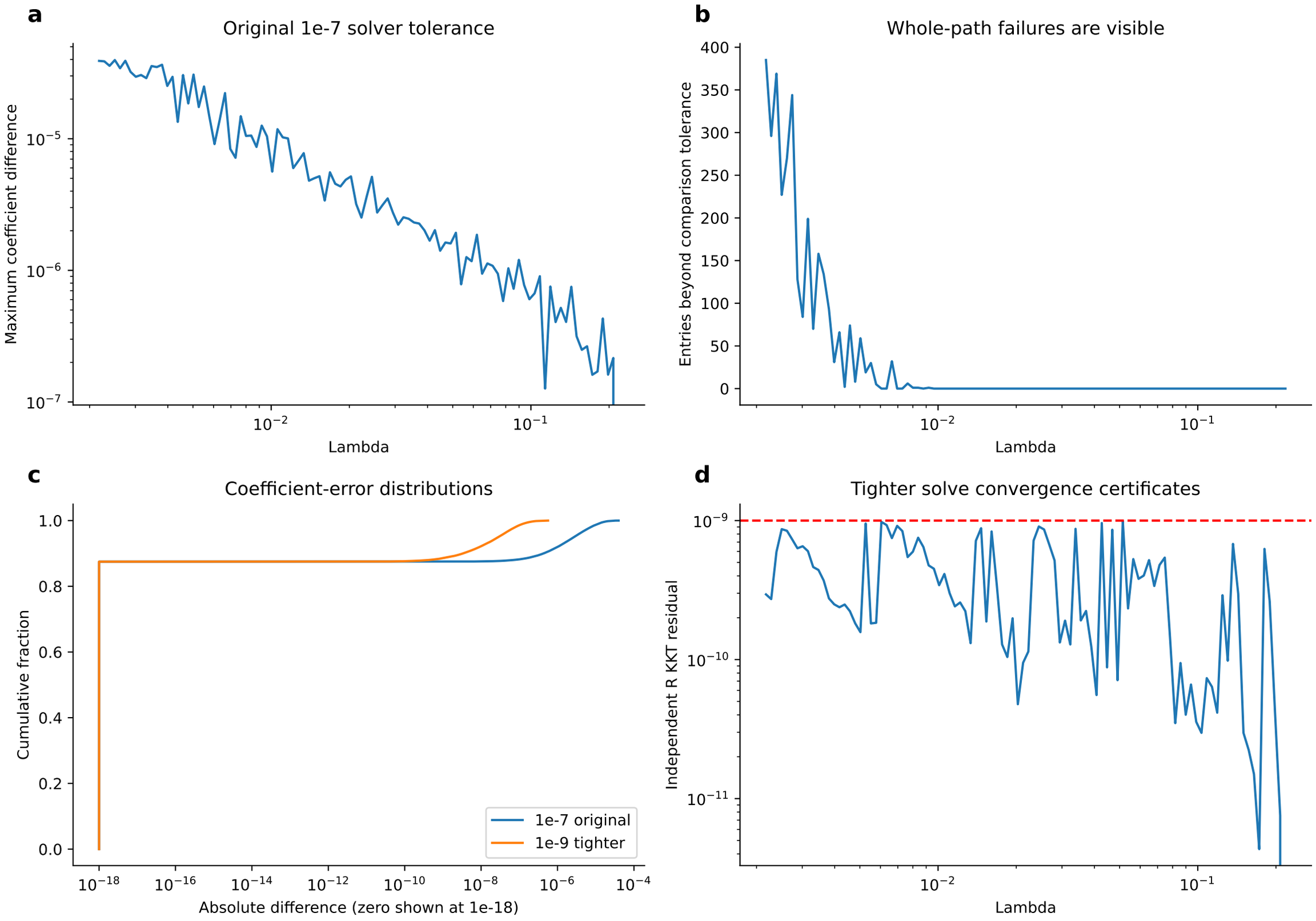
a
Original 1e-7 solver tolerance
Maximum coefficient difference
10^{-5}
10^{-6}
10^{-7}
10^{-2}
10^{-1}
Lambda
b
Whole-path failures are visible
Entries beyond comparison tolerance
400
350
300
250
200
150
100
50
0
10^{-2}
10^{-1}
Lambda
c
Coefficient-error distributions
Cumulative fraction
1.0
0.8
0.6
0.4
0.2
0.0
10^{-18}
10^{-16}
10^{-14}
10^{-12}
10^{-10}
10^{-8}
10^{-6}
10^{-4}
Absolute difference (zero shown at 1e-18)
1e-7 original
1e-9 tighter
d
Tighter solve convergence certificates
Independent R KKT residual
10^{-9}
10^{-10}
10^{-11}
10^{-2}
10^{-1}
Lambda